A whole message in light

How Holograms Work

Name _____ Date _____



Little Wonderer tilts a plain card with a shiny square on it. Inside the square, a tiny bird seems to sit behind the card. Tilt it left, and you see one side of the bird. Tilt it right, and you see the other side. Big Wonderer crouches low to look. How can a flat square hold something you can look around?



Light travels in waves, a little like water. Tap two fingers in a pan of water at once. Two sets of rings spread out and cross. Where two bumps meet, they stack into a taller bump. Where a bump meets a dip, they cancel into a calm line. This crossing is called interference. A hologram records it with light.



To make a hologram, a scientist splits one laser beam into two. One beam bounces off a seashell and onto the film. The other beam shines straight onto the film. The two beams cross, just like the two ripples. The film records bright and dark stripes, finer than a hair. It looks nothing like a shell!

hologram	a recording of light waves that shows depth when lit
interference	what happens when waves cross and stack up or cancel
laser	a very neat beam of light of one color
groove	a long, thin dip cut or pressed into a surface
reflection	light bouncing off a surface, like you in a window

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Here is a strange clue. Break a hologram plate in half, and each half still shows the whole shell! Light from every spot on the shell reached every part of the film. So every part holds the whole scene, seen from its own spot. The name hologram comes from Greek words for whole and writing.



Back at the table, the card stands under a lamp. Big Wonderer watches rainbow colors slide across it. The little one points at the bird. Now the guess changes. The bird is not painted on. The shiny foil is stamped with tiny grooves. They bend the lamplight back into the bird's light waves. Each eye gets a slightly different view, so the bird looks deep.



In movies, glowing pictures float in the air. Most stage versions are an old trick from 1862. A slanted sheet of clear glass reflects a lit figure, so it seems to hang in the air. A real hologram needs film or foil to bend the light. It also needs a very still table, because one wiggle can blur the stripes. Could light ever draw a bird in empty air?

1. What happens where two ripples cross?
2. What two beams meet on the film?
3. Why does half a hologram still show the whole shell?
4. Why does the bird on the card look deep?

Your turn

Write or dictate three sentences: what the ripples in the pan show, what is really on a hologram's film, and why the bird looks deep.

Example: Where ripples cross, bumps stack or cancel. The film holds tiny stripes. Grooves bend light, so each eye sees a different view.